

Belief-Based Sequential Decision Making for Financial Transaction Fraud Detection

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Abstract

Financial transaction fraud detection is a sequential decision problem under uncertainty: the true state of a transaction is hidden, the cost of a wrong decision is asymmetric, and a cautious agent may defer an irreversible action by pausing the transaction and collecting additional evidence. We present a research prototype of a transaction-decision agent that (i) summarizes a customer’s historical behavior into a lightweight profile, (ii) extracts rule-based evidence signals by comparing an incoming transaction to that profile, (iii) maintains an explicit Bayesian belief over the hidden legitimate/fraudulent state, and (iv) selects among APPROVE, HOLD, and STOP by minimizing expected cost, where HOLD triggers a customer-verification feedback loop that updates the belief before a final decision. We evaluate the agent on a frozen, ground-truth-blind benchmark of 35 adversarially constructed cases. The agent matches the expected decision on 22/35 cases (62.9%). We decompose all 13 mismatches into three categories: evidence dilution under conditional independence, decision-boundary calibration, and signals not yet modeled (transaction velocity, merchant-category familiarity, profile maturity). Using twin-case pairs, we empirically demonstrate that the agent’s decisions do not leak ground-truth labels. The system is implemented in pure Python (standard library only; 23 passing unit and integration tests).

1 Introduction

Fraud detection for financial transactions is dominated by machine-learning classifiers that map a transaction to a fraud score and then threshold the score [1; 4; 5]. Two properties of the problem are frequently underspecified in this paradigm. First, the cost of a mis-decision is strongly asymmetric and context-dependent: approving a fraudulent high-value transfer is far more costly than delaying a legitimate low-value purchase, and the two errors have different business consequences [3]. Second, fraud detection is inherently a *sequential* decision problem: the transaction arrives with incomplete information, and the agent may avoid an irreversible decision

by pausing the transaction and soliciting verification evidence from the customer.

This paper describes a research prototype — the *Transaction Agent* — that treats the problem as one of decision-making under uncertainty rather than of classification. The agent (i) builds a per-customer behavioral profile from historical transactions, (ii) extracts interpretable evidence signals by comparing the incoming transaction with the profile, (iii) updates an explicit Bayesian belief over the hidden state using a small, fully transparent likelihood table, and (iv) chooses among APPROVE, HOLD, and STOP by minimizing expected cost. HOLD is a first-class information-gathering action: it pauses the transaction, requests customer verification, and feeds the response back into the belief engine before a final decision.

The prototype avoids all learned components: there are no classifiers, no gradient updates, and no pretrained models. Every decision can be traced to a handful of explicitly stated probabilities and costs, which makes the system amenable to audit and to scientific study of its failure modes. We evaluate it on a frozen benchmark of 35 adversarially constructed scenarios in which the agent is kept strictly blind to ground truth. Our contributions are:

- A complete, standard-library-only implementation of a sequential Bayesian decision agent for transaction fraud, with HOLD as a first-class evidence-gathering action, validated by 23 automated tests.
- A ground-truth-blind evaluation protocol on a frozen 35-case benchmark, including *twin-case* pairs with identical evidence but opposite hidden states, used to empirically verify the absence of label leakage.
- A quantitative failure taxonomy of the 13 benchmark mismatches, isolating three concrete modeling gaps: conditional-independence dilution, decision-boundary calibration, and missing signal channels (velocity, merchant-category familiarity, profile maturity).

2 Related Work

Statistical fraud detection. Bolton and Hand [1] survey statistical methods for fraud detection and emphasize that fraud detection is a decision problem in which the cost structure of errors matters more than raw accuracy. Hand and Henley [2]

reach a similar conclusion for consumer credit scoring, treating scorecard thresholds as decision points that should reflect the economics of lending. Our work follows this decision-theoretic tradition rather than the accuracy-centric view of modern classifier benchmarks.

Cost-sensitive learning. Bahnsen et al. [3] provide the closest methodological ancestor: they formulate credit-card fraud detection as a Bayes minimum-risk problem in which classification is performed by minimizing the expected misclassification cost instead of the error rate. We adopt the same principle but replace their expensive per-model cost matrix estimation with an explicit, interpretable cost model and extend the single decision into a *sequential* loop in which one action (HOLD) yields additional evidence.

Modern fraud-detection benchmarks. Large public data efforts such as the IEEE-CIS Kaggle competition [6] and the reproducible Fraud-Detection Handbook [5] provide transactional data and accurate/recalibrated classifier baselines. These datasets are valuable for future validation of the agent’s belief engine, but their evaluation protocol is classification-oriented; they do not model deferred decisions or information acquisition, which are central to our setting.

Uncertainty in agents. The emerging literature on uncertainty quantification in interactive LLM agents studies how an agent’s estimate of its own uncertainty changes as it interacts and gathers information. Our verification loop is a concrete, non-learned instance of this principle: the agent tracks a calibrated probability and chooses to gather more evidence only when its belief falls in an ambiguous region. This connects the classical decision-theoretic machinery with the emerging “act-to-resolve-uncertainty” view of agents.

3 Problem Statement

A transaction T arrives with observable features: amount A , merchant M , timestamp τ , and location L . The *hidden state* $H \in \{\text{LEGIT}, \text{FRAUD}\}$ is unobserved. The agent also has access to the customer’s historical transactions, from which a profile is derived. The agent must select one of three actions:

- APPROVE: release the transaction immediately;
- HOLD: pause the transaction and request customer verification;
- STOP: terminate the transaction immediately.

Each action has a known cost $C(a \mid H)$ for each hidden state. The agent’s goal is to minimize the expected cost given its current belief. This is a two-stage sequential decision: if HOLD is chosen, a verification response V arrives and the belief is updated before the agent commits to a terminal action. Because APPROVE and STOP are irreversible, the problem is a degenerate (two-stage) partially observable Markov decision process, in which the single “act” that changes the information state is the verification solicitation.

4 The Transaction Agent

The agent’s pipeline is summarized in Algorithm 1. Its modules are cleanly separated: profiling, evidence extraction, belief updating, decision making, and verification. All param-

eters are explicit and modular; each decision can be reproduced by hand from the tables in this paper.

4.1 Behavioral Profiling

Historical transactions for a customer c are summarized into a profile S_c :

- *Typical amount*: the median of all non-missing historical amounts;
- *Common merchants, typical locations, and typical hours*: the sets of merchants, cities, and hour-of-day values appearing in the customer’s history, in order of first appearance.

Missing values in the historical records are handled gracefully: a customer with no usable amounts simply has no typical amount. In our experiments the history consists of 200 synthetic records across 10 customers (C001–C010).

4.2 Evidence Extraction

Four rule-based checkers compare the incoming transaction with S_c and emit one evidence signal each:

- *Amount*: if $A > 1.5 \times \text{median}$, emit `large_amount` (fraud-supporting); otherwise `amount_consistent`.
- *Merchant*: if $M \notin \text{common merchants}$, emit `new_merchant`; otherwise `merchant_consistent`.
- *Location*: if $L \notin \text{typical locations}$, emit `unusual_location`; otherwise `location_consistent`.
- *Time*: if $\tau.\text{hour} \notin \text{typical hours}$, emit `unusual_time`; otherwise `time_consistent`.

Any missing feature emits a neutral `*unknown` signal that leaves the belief unchanged. This keeps the system robust to incomplete transaction data.

4.3 Bayesian Belief Updating

The agent maintains a belief (p_L, p_F) over the hidden state, initialized with the prior

$$p_L^{(0)} = 0.90, \quad p_F^{(0)} = 0.10. \quad (1)$$

Each evidence signal E updates the belief by Bayes’ rule:

$$p_F \leftarrow \frac{p_F P(E \mid \text{FRAUD})}{p_F P(E \mid \text{FRAUD}) + p_L P(E \mid \text{LEGIT})}. \quad (2)$$

Under the conditional-independence assumption, evidence arriving sequentially is applied multiplicatively. In odds form the update is compact:

$$O(F \mid E_1 \dots E_n) = O^{(0)}(F) \prod_{i=1}^n \Lambda(E_i), \quad (3)$$

where each signal contributes a likelihood ratio

$$\Lambda(E_i) = \frac{P(E_i \mid \text{FRAUD})}{P(E_i \mid \text{LEGIT})}, \quad (4)$$

and $O(F) = p_F/p_L$. Table 1 lists the prototype likelihood table, which is fully explicit and auditable. Consistency signals carry likelihood ratios below 1 (they push the belief toward legitimate), anomaly signals carry ratios above 1, and the two verification signals are the strongest evidence available: a customer denial has ratio 19, a confirmation ratio 0.053.

Algorithm 1 Transaction decision loop

Input: Customer history $\mathcal{H}_c$, incoming transaction T
Parameter: Prior $(0.90, 0.10)$; LR s (Tab. 1); costs (Tab. 2)
Output: Final action $a^* \in \{\text{APPROVE}, \text{STOP}\}$

- 1: Build profile S_c from $\mathcal{H}_c$ (median amount, merchants, hours).
- 2: Extract evidence $\{E_1, \dots, E_4\}$ by comparing T with S_c .
- 3: Initialize belief $b \leftarrow (0.90, 0.10)$.
- 4: **for** each E_i **do**
- 5: Update b with Bayes' rule (Eq. (2)).
- 6: **end for**
- 7: Compute $\mathbb{E}[C | a]$ for all a (Eq. (5)).
- 8: **if** $\arg \min_a \mathbb{E}[C | a] = \text{HOLD}$ **then**
- 9: Request customer verification; obtain response V .
- 10: Convert V to verification evidence (customer_confirms / customer_denies).
- 11: Update b again; recompute expected costs.
- 12: Return the new argmin (over APPROVE / STOP).
- 13: **else**
- 14: Return the argmin action directly.
- 15: **end if**

4.4 Expected-Cost Decision Making

The expected cost of action a under belief (p_L, p_F) is

$$\mathbb{E}[C | a] = p_L C(a | \text{LEGIT}) + p_F C(a | \text{FRAUD}). \quad (5)$$

The default cost model (Table 2) sets approving a fraudulent transaction at cost 10, stopping a legitimate one at cost 8, and treating verification (HOLD) as cheap and symmetric (2), reflecting low customer friction for an out-of-band confirmation.

Theorem 1 (Decision regions). *Under the cost model of Table 2, the expected-cost-minimizing action is a function of p_F alone:*

$$a^*(p_F) = \begin{cases} \text{APPROVE} & p_F < 0.2, \\ \text{HOLD} & 0.2 \leq p_F \leq 0.75, \\ \text{STOP} & p_F > 0.75. \end{cases} \quad (6)$$

Proof. From Eq. (5): $\mathbb{E}[C | \text{APPROVE}] = 10p_F$, $\mathbb{E}[C | \text{HOLD}] = 2$, $\mathbb{E}[C | \text{STOP}] = 8(1 - p_F)$. Comparing the first two, APPROVE is optimal iff $10p_F < 2$, i.e. $p_F < 0.2$. Comparing the last two, HOLD beats STOP iff $2 < 8(1 - p_F)$, i.e. $p_F < 0.75$. $\square$

A subtle discovery from the prototype's development is worth reporting here: the agent's *first* cost model set HOLD at (legit = 6, fraud = 4), which makes HOLD dominated — its expected cost exceeds both APPROVE and STOP at every belief point in $[0, 1]$. HOLD was mathematically inadmissible, a fact that was invisible in an accuracy-framed classifier but immediately surfaced in the expected-cost formulation. The default model of Table 2 was adopted specifically so that the three decision regions of Theorem 1 are non-degenerate.

Evidence signal	$P(E \text{FRG})$	$P(E \text{LGT})$	Λ
large_amount	0.30	0.20	1.50
amount_consistent	0.70	0.80	0.875
new_merchant	0.30	0.15	2.00
merchant_consistent	0.70	0.85	0.824
unusual_location	0.30	0.10	3.00
location_consistent	0.70	0.90	0.778
unusual_time	0.40	0.10	4.00
time_consistent	0.60	0.90	0.667
customer_confirms	0.05	0.95	0.053
customer_denies	0.95	0.05	19.00

Table 1: Prototype likelihood table with ratios $\Lambda = P(E | \text{FRAUD})/P(E | \text{LEGIT})$. Consistency signals are complements of anomaly signals; “unknown” signals are neutral and omitted.

Hidden state	APPROVE	HOLD	STOP
LEGIT	0.0	2.0	8.0
FRAUD	10.0	2.0	0.0

Table 2: Default cost model. Asymmetric costs make the optimal action a pure function of p_F (Theorem 1).

4.5 Verification as Information Gathering

If the initial decision is HOLD, the agent creates a pending verification request, records the customer's response, converts it into verification evidence (customer_confirms or customer_denies), updates the belief once more, and re-runs expected-cost minimization over the two terminal actions. Because a denial carries likelihood ratio 19, a single failed verification can push a p_F in the ambiguous $[0.2, 0.75]$ band above the 75% STOP threshold; a confirmation pushes it far below 20%. This realizes, in explicit probabilistic form, the principle that a *reversible* action at a 50/50 belief is often the cheapest choice.

5 Evaluation

5.1 Benchmark Design and Ground-Truth Blindness

The benchmark file used throughout this evaluation¹ is *frozen* and read-only: it contains 35 scenario cases (TC01–TC35) and must not be regenerated. Each case specifies the observed features (amount, merchant, time, location), a natural-language description of the customer's historical behavior, explicit evidence notes, and — hidden from the agent — the ground-truth state (22 legitimate, 13 fraudulent) and the human-expected action (12 APPROVE, 15 HOLD, 8 STOP).

The evaluation runner computes the agent's decision from feature columns only. Ground truth is read solely for post-hoc comparison, after the decision is finalized. The set deliberately contains five *twin pairs* — cases with identical evidence but opposite hidden states (e.g., TC04/TC05, TC10/TC11) — so that any leakage of ground truth into the decision would be directly observable as a disagreement between twins.

¹transaction_agent_test_cases_v2.csv in experiments/.

Case	Evidence summary	p_F	Agt./Exp.	Match
TC01	4 consistent	0.040	A/A	✓
TC02	4 anomalies	0.800	S/S	✓
TC04	large amt + 3 ok	0.066	A/H	×
TC07	unusual time + 3 ok	0.199	A/H	×
TC08	3 anom. + 1 ok	0.509	H/S	×
TC17	2 anom. (time+loc)	0.490	H/H	✓
TC20	3 txs in 6 min	0.092	A/S	×
TC25	unfamiliar merc. (cat.)	0.092	A/H	×
TC33	extreme amt + 4 anom.	0.800	S/S	✓
TC35	sparse profile (<5 txs)	0.092	A/H	×

Table 3: Representative benchmark cases. p_F is the posterior fraud probability after belief updating; A/H/S denote APPROVE/HOLD/STOP; Match compares the agent’s decision with the human-expected action.

5.2 Results

The agent selected APPROVE 21 times, HOLD 9 times, and STOP 5 times, matching the expected action on 22/35 cases (62.9%). Table 3 shows representative cases. Consistent transactions (TC01, TC19) are approved with posterior fraud probability near 4%; fully anomalous transactions (TC02, TC23, TC33) are stopped with $p_F = 80\%$; and genuinely ambiguous two-anomaly cases (TC17, TC18) land in the HOLD band, exactly where the cost model intends.

5.3 Failure Taxonomy

All 13 mismatches fall into three categories, each with a distinct mechanism:

Category A: Conditional-independence dilution (7 cases). TC04, TC05, TC10, TC11, TC13, TC22, TC07. A single moderate anomaly (e.g. a large amount) combined with three routine consistent signals is diluted by the multiplicative independence update. In TC04/TC05 the product of likelihood ratios is $1.5 \times 0.824 \times 0.778 \times 0.667 = 0.642$, driving p_F down from 10% to 6.6% below the 20% HOLD threshold, even though the amount is $4\times$ the historical maximum. TC07 reaches $p_F = 19.94\%$, missing the boundary by 0.06 points. This is a modeling limitation, not a software bug: naive conditional independence lets many “routine” signals cancel an extreme outlier.

Category B: Decision-boundary behavior (2 cases). TC08, TC14. The benchmark expected STOP; the agent chose HOLD with $p_F \in [0.40, 0.51]$. Because STOP costs 8 against a legitimate transaction, its expected cost is $8(1 - p_F) \geq 3.93$, which exceeds the constant cost 2 of HOLD. Under the stated cost model, pausing a 50/50 transaction for verification is strictly cheaper than an irreversible cancellation — the agent’s choice is the rational one under Theorem 1. The disagreement reveals a normative difference in how aggressively fraud should be terminated, i.e. a cost-calibration question, not a reasoning error.

Category C: Missing signal channels (4 cases). TC20 (three transactions within six minutes — transaction *velocity* is not represented in the four single-transaction checkers), TC25/TC26 (an unfamiliar merchant within a familiar spending category — merchant familiarity is currently binary), and

TC35 (a profile with fewer than five prior transactions — profile *maturity* is not modeled). Each of these is a deliberately deferred feature whose signal channel does not yet exist.

5.4 Twin-Pair Probe for Label Leakage

For every twin pair (TC04/TC05, TC10/TC11, TC17/TC18, TC25/TC26, TC31/TC32), the agent produced the identical decision for both members, despite opposite ground-truth states. Since the two members share evidence and differ only in the hidden label, identical outputs are consistent with the agent operating purely on features. This serves as an empirical control that the ground-truth columns do not influence the decision.

6 Discussion

The prototype’s value is primarily investigative. Three conclusions emerge. First, an explicit belief state makes otherwise subtle modeling assumptions visible. The conditional-independence assumption is attractive for its simplicity and its closed-form updates (Eq. (3)), but the benchmark shows it over-weights repeated routine evidence against a single extreme anomaly. Second, expected-cost decision making surfaces normative tension that accuracy metrics hide: the TC08/TC14 disagreements are not errors but a stance on the economics of false stops, and the history of the cost model shows how easily a classifier designer can make a reasonable action (HOLD) structurally unused. Third, the twin pairs and the frozen benchmark provide a reusable evaluation protocol for future signal channels.

The system has deliberate limitations. All probabilities, likelihoods, and costs are synthetic research assumptions, not calibrated banking statistics; parameters must be estimated from real data before deployment. The evidence set is small (four channels) and single-transaction; and the verification loop assumes a reliable response channel. Deployment concerns (timing, fraud rings, identity-verification cost) are out of scope for this prototype.

7 Future Work

The benchmark failures define the research agenda directly. (i) *Non-linear evidence combination*: extreme individual anomalies should resist dilution, e.g. by capping the strength of routine signals or introducing a generalized aggregation rule; TC07’s 0.06-point miss indicates that even the boundary behavior is fragile. (ii) *New signal channels*: transaction velocity, merchant-category familiarity, and profile-maturity estimates should become first-class evidence channels with their own likelihood entries. (iii) *Cost calibration*: the HOLD/STOP boundary should be studied against real false-positive and false-negative economics, and the cost model should be parameterized per transaction context (e.g., low versus high denomination). (iv) *Empirical validation*: the agent’s belief engine and likelihood table should be confronted with real transaction data such as the IEEE-CIS dataset [6] after estimating parameters from observed frequencies. We also plan to study the connection between the verification loop and uncertainty-driven information acquisition in agents.

Ethical Statement

This paper reports a research prototype evaluated only on synthetic data. No real customer data was used, and no deployment or financial decisions were influenced by the system. The benchmark cases are fabricated and carry no personally identifiable information.

References

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